Family Name _____

First Name _____

Student Number |__|__|__|__|__|__|__|__|

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No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Computer Science and Software Engineering

EXAMINATION

End-of-Year Examinations, 2023

ENCE260-23S2 (C) Computer Systems

Examination Duration: 120 minutes

Exam Conditions:

Closed Book: Students may bring in writing implements only, no materials.

Calculators with a 'UC' sticker approved.

Materials Permitted in the Exam Venue:

None

Materials to be Supplied to Students:

1 x Write-on exam paper

Instructions to Students:

ALL questions are COMPULSORY

There are FOUR questions in total

Maximum TOTAL marks 100

For Examiner Use Only

Question	Mark

Total _____

INSTRUCTIONS

You are required to complete this examination by writing your answers into this booklet. Show your working as you complete this examination booklet, so that you can more easily receive partial credit.

If you need more space for your answers, several extra pages can be found at the end of this booklet. Make a note indicating when your work is continued on an extra page.

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Exam starts on Page 4

Q1. **(a)** Logic functions and design:

(i) **[2 marks]**

Using the following truth table, represent the logic function S as a sum-of-minterms, such that $S = (A \oplus B) \cdot C$.

A	B	C	
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(ii) **[2 marks]**

Using the rules of Boolean algebra, **simplify** the "sum-of-minterms" expression for S to use only AND, OR, and NOT logic functions.

(iii) [2 marks]

Sketch an implementation of your simplified Boolean expression for S using only AND, OR, and NOT logic gates.

(b) You are designing the control aspect of a CPU, and you know that you need all of the subcomponents shown in Figure 1.

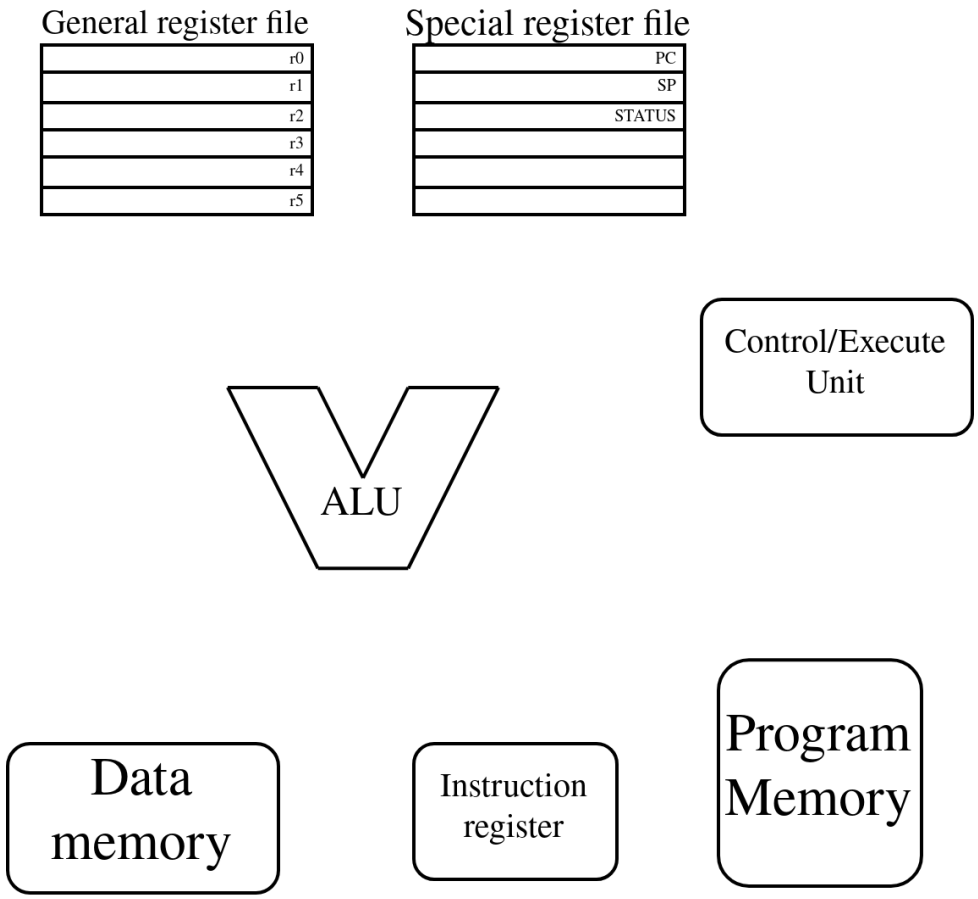


Figure 1: Block diagram showing the subcomponents required for a CPU.

Unfortunately, the diagram that you found does not show how everything is connected.

(i) [3 marks]

Draw arrows between appropriate components in Figure 1 showing all of the required data transfer and/or control connections between all of the shown subcomponents.

(ii) [2 marks]

Show the purpose of the connections that you have identified, by **labelling all of the arrows** you have drawn with **either**: C for 'C'ontrol, OR D for 'D'ata.

The ATMega32u2 is a good example to use to learn about computer architecture, and you find its block diagram, shown in Figure 2.

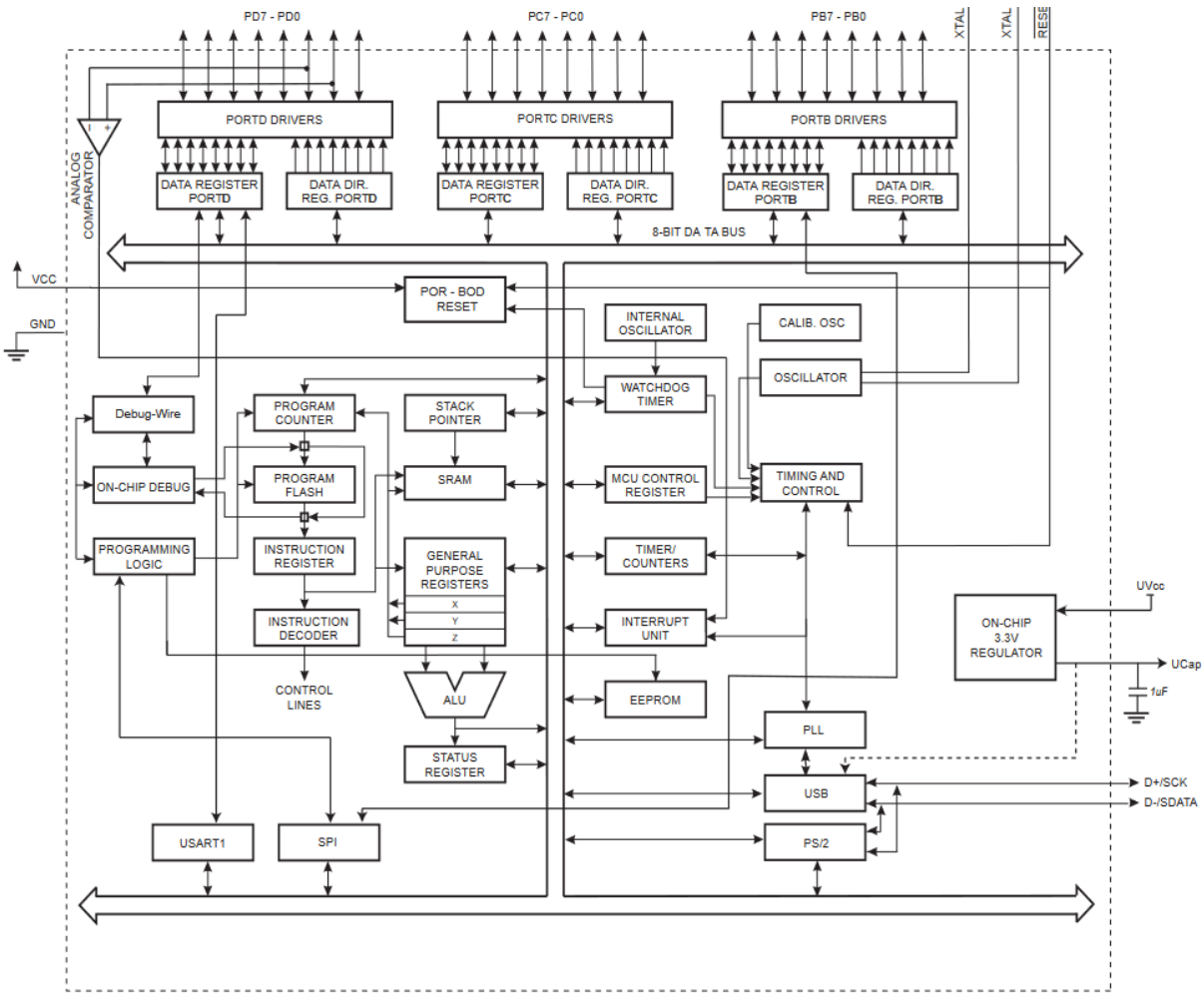


Figure 2: Block diagram of the ATMega32u2.

(iii) [4 marks]

On the diagram in Figure 2, use letters A through F to label the component that best defines each the following:.

A) Something that would interface with an LED.

B) Somewhere the direct result of an arithmetic operation would be stored.

C) A peripheral used to wait $500\ ms$.

D) The means by which data is transferred throughout the microcontroller

E) The control and execute unit.

(iv) [2 marks]

F) The component(s) which proves this is a Harvard arcitecture, OR circle: **NONE**.

(c) Table 1 is the truth table for an individual Full Adder circuit which performs addition on 1-bit numbers. The 4-bit Ripple-Carry Adder shown in Figure 3 uses four Full Adder circuits to perform addition on 4-bit binary numbers.

Table 1: Truth table for a Full Adder.

Inputs			Outputs	
A	B	C_{in} (carry in)	S (sum)	C_{out} (carry out)
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	0	0	1
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	1

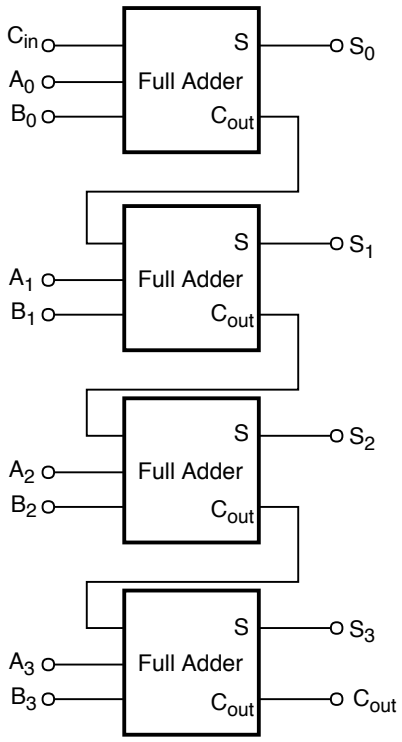


Figure 3: 4-bit Ripple-Carry Adder for context.

Consider now an 8-bit ripple counter, as would be found in an 8-bit ALU.

(i) [1 mark]

The Ripple-Carry Adder is to be used to analyse the base-10 numbers A = 11 and B = 25. Write the 8-bit binary sequences, $A_7A_6A_5A_4A_3A_2A_1A_0$ and $B_7B_6B_5B_4B_3B_2B_1B_0$, that represent the inputs to the Ripple-Carry Adder as binary sequences.

(ii) [2 marks]

The input B is to be inverted prior to input to the adder, and $C_{in} = 1$.

State what mathematical operation this will result in two's complement representation, **AND** state what type of logic gate can be used to perform an electronically-controlled bit-wise inversion.

(iii) [3 marks]

Complete the operation, by performing a bitwise addition of: A, the inverted B, and 1 from $C_{in} = 1$.

(iv) [2 marks]

Interpret your answer from **(iii)** in two's complement, **AND** state two other things the binary sequence could represent. [if you have no answer from above use `0b1110 1100`.]

Q2. **(a)** Memory is one of the critical parts of any computer device. One of the base units of memory is the flip-flop, for example that shown in Figure 4.

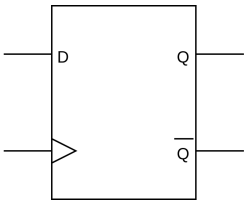


Figure 4: A flip-flop

(i) **[2 marks]**

State two functions that memory must perform, and briefly discuss how a flip-flop achieves these required functions.

Because of the inherent ability of a flip-flop to remember, they provide the ability to implement sequential logic.

(ii) **[1 mark]**

State the difference between sequential logic and combinational logic.

An example of a sequential logic circuit is shown in Figure 5.

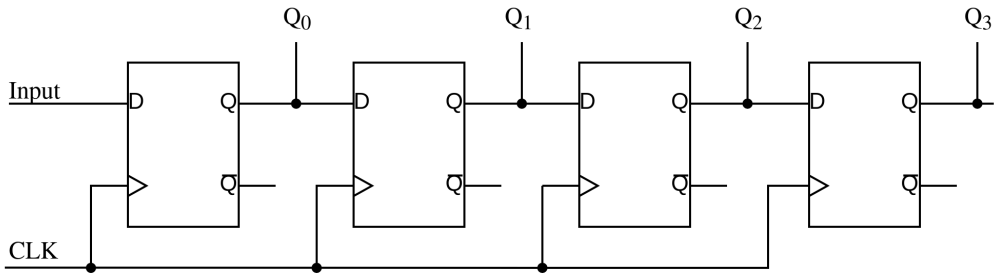


Figure 5: An example 4-bit sequential logic circuit.

(iii) **[1 mark]**

Name the sequential logic circuit shown in Figure 5.

The electrical signals provided to the `input` and the `CLK` signals are shown in Figure 6.

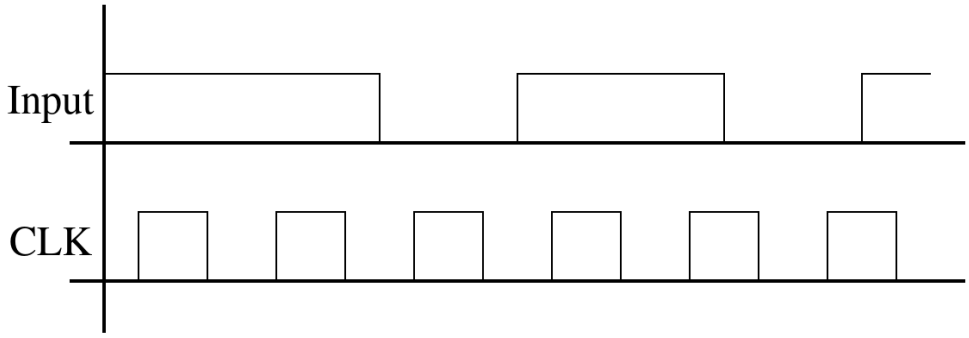


Figure 6: Electrical waveforms provided to the sequential logic circuit.

(iv) [2 marks]

Determine each of the signals $Q_0Q_1Q_2Q_3$ **after the end of the entire waveforms** shown in Figure 6.

(b) You are designing a system with the ATMega32u2, and want the ability to interface with several buttons and LEDs. Because you have not completely decided on which components are going to be connected to which pins, programmable bidirectional GPIO pins are required, the schematic for which is shown in Figure 7.

Figure 12-2. General Digital I/O(1)

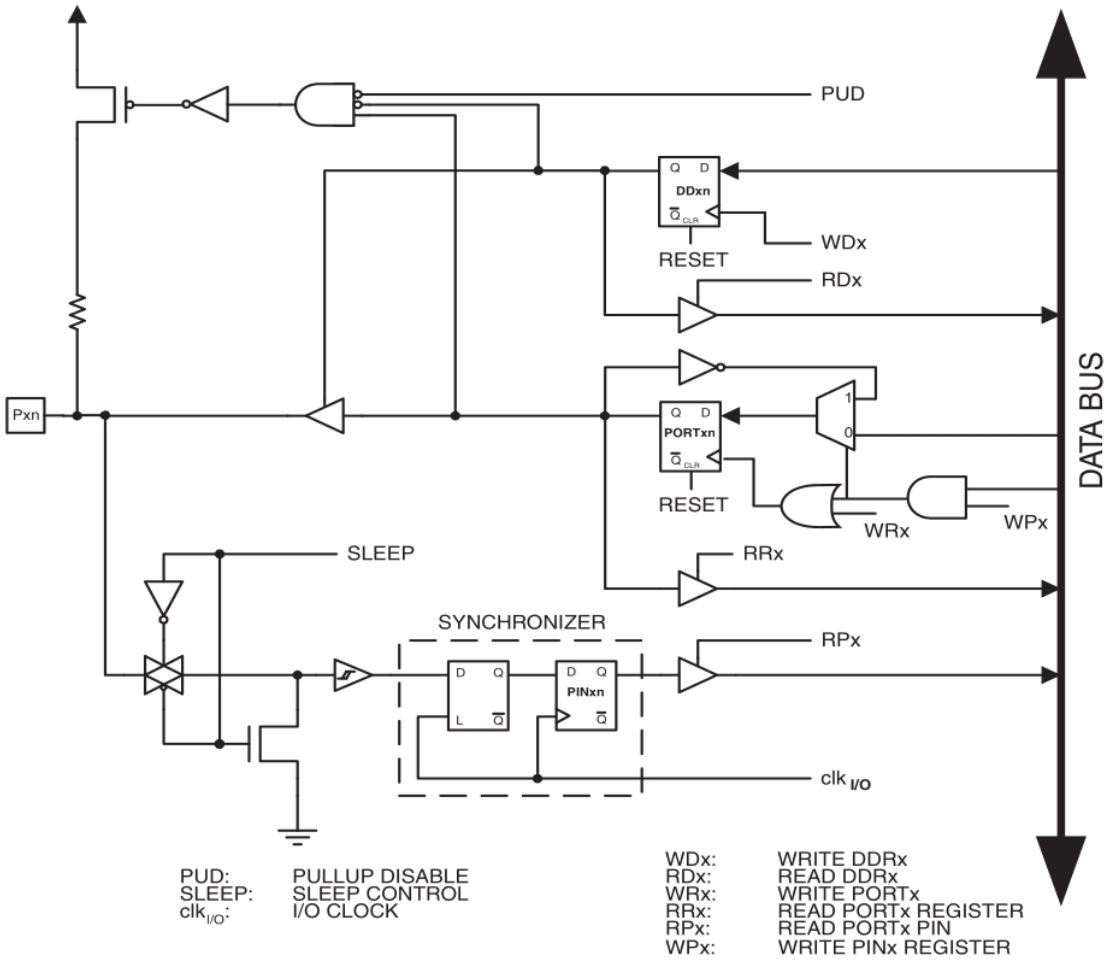


Figure 7: Diagram of the internal GPIO circuitry of the ATMega32u2.

(i) [5 marks]

State the four possible operational states that enable bidirectional digital operation of the pin `Pxn`, and, with reference to Figure 7, state how this bidirectionality is achieved. In your answer, state the key registers that are used to manipulate the pin.

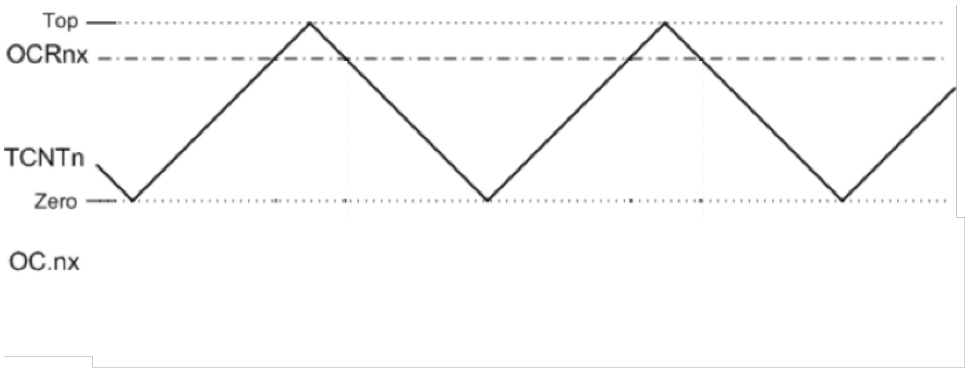


Figure 8: Diagram that illustrates how to generate a pulse width modulation output from a timer.

You are creating a PWM signal to power an LED, and you know that a comparison is made with a value increasing or decreasing at a contstant rate, such as that shown in Figure 8. You know that you can introduce a `uint8_t` value, and increment it at a certain period (in software), or you can use the built-in timer/counter peripheral (in hardware).

(ii) [3 marks]

State which is the more robust solution, and justify your answer.

(iii) [2 marks]

You know that if you connect your PWM output to an LED directly, you may damage the microcontroller. State why this is, and suggest two methods to avoid damage.

(c) An AVR instruction for the ATMega32u2 is shown in Table 2.

Table 2: BRNE — branch if not equal — opcode specification

Syntax	Operands	Operation
`brne k`	$-64 \leq k \leq 63$	Result of CP, CPI, SUB, or SUBI $\neq 0$?
Control registers	Opcode	Flags
PC $\leftarrow$ PC + k + 1 (if Z = 0) PC $\leftarrow$ PC + 1 (if Z = 1)	1111 01kk kkkk k001	None

(i) [3 marks]

State what type/class of instruction is the instruction 'branch if not equal', and how does it achieve this functionality?

Another AVR instruction is shown in Table 3

Table 3: CPI — Compare immediate — opcode. Note: d has an offset of +16

Syntax	Operands	Operation
`cpi RD,K`	$16 \leq d \leq 31, 0 \leq K \leq 255$	Rd - K
Control registers	Opcode	Flags
PC $\leftarrow$ PC + 1	0011 KKKK dddd KKKK	H,S,V,N,Z,C

(ii) [2 marks]

The CPI instruction uses immediate addressing. Explain what the term 'immediate addressing' means, and name two other addressing modes.

Table 4: Example of some AVR instructions as compiled for the ATMega32u2.

PC	opcode (hex)	instruction
...		
0x0008:	3c 20	cpi r18 0xc0;
0x0009:	f4 a1	brne, 20
...		

Table 5: Example register values.

Register	Value
...	
R16	01
R17	00
R18	c0
R19	1a
R20	3c
R21	c2
R22	00
R23	f0
...	

(iii) [3 marks]

In words, describe the action of the two instructions shown in Table 4 given register contents shown in Table 5. (You do not have to logically/numerically solve anything here).

(d) [1 mark]

How is every piece of information within computers stored?

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Q3. Embedded Systems 1

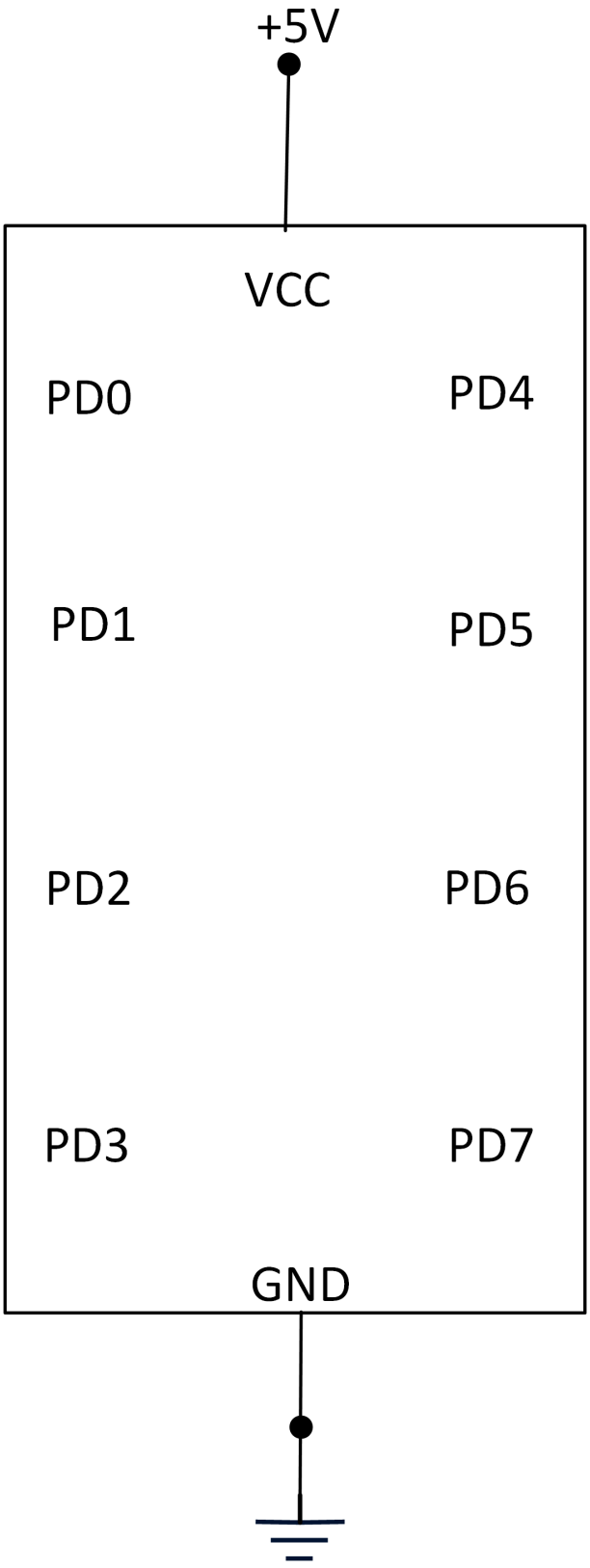


Figure 9: Schematic showing an ATmega32u2 microcontroller to be filled in for Q3(a).

(a) Port D of an ATmega32u2 micrcocontroller is connected to one switch and five LEDs (L1, L2, L3, L4, L5), but the input/output pins for these are unknown. Three of the five LEDs are each connected to one PIO pin only. Two of the five LEDs are each connected across two PIO pins. With the following information, draw the switch, LEDs and any required associated electronics (resistors etc) into Figure 9. Note that there is an extra copy of Figure 9 in the Extra Space at the end of the booklet if you need to redo Question 3.

(i) [3 marks]

The following line evaluates to `True` when the switch is pushed:

```
if ((PIND & (1<<3))==0)
```

Draw the switch and any required electronics for correct operation into Figure 9.

(ii) [2 marks]

What **decimal** value should be stored in the Data Direction Register for Port D (DDRD)?

(iii) [2 marks]

L1 is an active low LED and is toggled by:

```
PORTD ^= (1<<0);
```

Draw and label L1 and any required electronics for correct operation into Figure 9.

(iv) [2 marks]

L2 and L3 are both active high LEDs and are the only LEDs turned on by:

```
PORTD = 0x C1;
```

Draw L2 and L3 and any required electronics for correct operation into Figure 9. (It does not matter which one you label L2 or L3).

(v) [3 marks]

LEDs L4 and L5 both connect across two adjacent PIO pins. L4 and L5 are the only LEDs turned on by:

```
PORTD = 35;
```

Draw L4 and L5 and any required electronics for correct operation into Figure 9. (It does not matter which one you label L4 or L5).

Question 3 continues on p22

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(b) Faye and Milo are writing a program `beep1` on the UCFK4. The file dependency graph for their program is shown in Figure 10. Note that all `.c` files also depend on `system.h` but these are not shown in Figure 10 for clarity. Faye has started writing the Makefile in Listing 1. Both Faye and Milo have this version of `beep1` in their local repositories and in the remote repository.

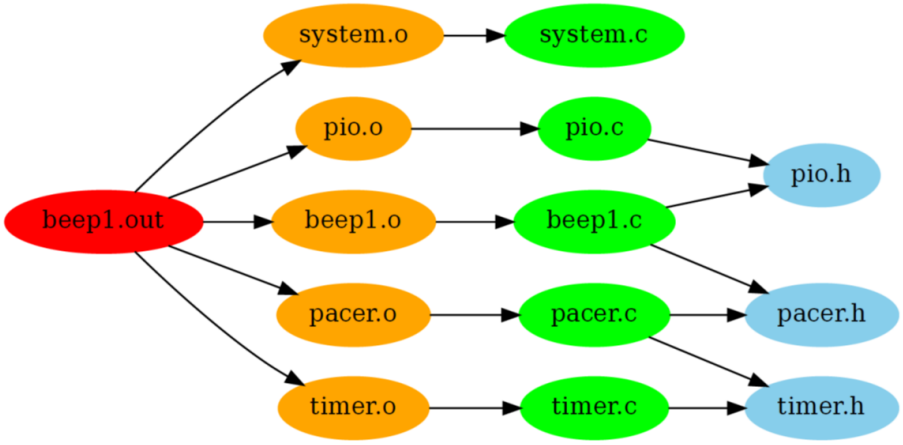


Figure 10: File dependency graph for Q3(b).

(i) [8 marks]

Complete the Makefile in Listing 1 for Faye so that the `beep1` program is loaded onto the microcontroller when `make program` is called. You should include `system.h` in your compilation steps. You may assume all the necessary files are stored in the same directory as `beep1.c` as has been done in Line 8 of Listing 1.

Listing 1: Makefile for Question 3(b).

```
CC = avr-gcc
CFLAGS = -mmcu=atmega32u2 -Os -Wall
OBJCOPY = avr-objcopy
DEL = rm

all: beep1.out

beep1.o: beep1.c pio.h system.h pacer.h
    $(CC) -c $(CFLAGS) $< -o $@









.PHONY: clean
clean:
    -$(DEL) *.o *.out *.hex

.PHONY: program
program: beep1.hex
    dfu-programmer atmega32u2 erase; dfu-programmer \
        atmega32u2 flash beep1.hex; \
            dfu-programmer atmega32u2 start
```

(ii) [2 marks]

Of the file types used to produce the `beep1` program, which should be included in a version control system such as git? **Why?**

(iii) [1 mark]

Faye updates the remote repository with your suggested changes. Use the following box for the git command(s) that Milo needs to write so that his local repository includes these changes from Q3(b)i.

(iv) [2 marks]

Milo notices a bug in this new version and wishes to remove Faye's changes. Use the following box to write the git command(s) to modify Milo's local repository back to the previous version.

Q4. Embedded Systems 2

(a) Data is transmitted serially from one UART to another. Assume the UARTs have been configured for 7-bit data, the MSB is transmitted first, the idle state is high, and there are two stop bits, one start bit, and an even parity bit. Answer the following:

(i) [4 marks]

Sketch the waveform that should be transmitted from the UART to transmit the ASCII character `Y`. An ASCII table is shown in Table 6.

(ii) [2 marks]

If it takes $1.173\,\mathrm{s}$ to transmit 1024 characters, what is the Baud rate?

Table 6: The ASCII character table for Q4(a).

0	NUL	32		64	@	96	‘
1	^A	33	!	65	A	97	a
2	STX	34	"	66	B	98	b
3	ETX	35	#	67	C	99	c
4	^D	36	$	68	D	100	d
5	^E	37	%	69	E	101	e
6	^F	38	&	70	F	102	f
7	BELL	39	’	71	G	103	g
8	BS	40	(	72	H	104	h
9	TAB	41	)	73	I	105	i
10	LF	42	*	74	J	106	z
11	VT	43	+	75	K	107	k
12	FF	44	,	76	L	108	l
13	CR	45	-	77	M	109	m
14	^N	46	.	78	N	110	n
15	^O	47	/	79	O	111	o
16	^P	48	0	80	P	112	p
17	XON	49	1	81	Q	113	q
18	^R	50	2	82	R	114	r
19	XOFF	51	3	83	S	115	s
20	^T	52	4	84	T	116	t
21	^U	53	5	85	U	117	u
22	^V	54	6	86	V	118	v
23	^W	55	7	87	W	119	w
24	^X	56	8	88	X	120	x
25	^Y	57	9	89	Y	121	y
26	^Z	58	:	90	Z	122	z
27	ESC	59	;	91	[	123	{
28	FS	60	<	92	\	124	|
29	GS	61	=	93	]	125	}
30	RS	62	>	94	^	126	~
31	US	63	?	95	_	127	DEL

(b) Nate and Penny have written the C program `flasher.c` in Listing 2 for the ATmega32u2 microcontroller. The TCCR1B configuration register is shown in Figure 11, and the TCNT1 register in Figure 12. LED1 is in an active high configuration. Given a CPU frequency of 6 MHz, answer the following.

Listing 2: Source listing for Q4(b)

```
#include "pio.h"
#include "system.h"

void tempo (uint16_t ticks) {
    while (TCNT1 < ticks)
        continue;
    TCNT1 = 0;
}

void flasher(uint16_t alpha, uint16_t beta, uint16_t gamma)
{
    TCCR1B = 0x04;
    TCNT1 = 0;

    while (gamma--)    {

        pio_output_high(LED1);
        tempo(alpha+beta+gamma*(alpha/beta));
        pio_output_low(LED1);
        tempo(alpha+beta+gamma);

    }
}

int main (void)
{
    system_init ();
    pio_config_set(LED1, PIO_OUTPUT_LOW);
    flasher(30000,40000,1000);

    return 0;
}
```

Bit	7	6	5	4	3	2	1	0	
(0x81)	ICNC1	ICES1	–	WGM13	WGM12	CS12	CS11	CS10	TCCR1B
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Clock Select Bit Description

CSn2	CSn1	CSn0	Description
0	0	0	No clock source. (Timer/Counter stopped)
0	0	1	$clk_{I/O}$/1 (No prescaling
0	1	0	$clk_{I/O}$/8 (From prescaler)
0	1	1	$clk_{I/O}$/64 (From prescaler)
1	0	0	$clk_{I/O}$/256 (From prescaler)
1	0	1	$clk_{I/O}$/1024 (From prescaler)
1	1	0	External clock source on Tn pin. Clock on falling edge
1	1	1	External clock source on Tn pin. Clock on rising edge

Figure 11: TCCR1B register bits for Q4(b).

Bit	7	6	5	4	3	2	1	0	
(0x85)	TCNT1[15:8]								TCNT1H
(0x84)	TCNT1[7:0]								TCNT1L
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Figure 12: TCNT1 register bits for Q4(b).

(i) [2 marks]

What is the purpose of the `tempo` function in Listing 2?

(ii) [2 marks]

What is the downside of the `tempo` function in Listing 2? What alternative could be employed to overcome this problem?

(iii) [2 marks]

What is the timing resolution of the timer in Listing 2?

(iv) [1 mark]

How many times will the LED flash in Listing 2?

(v) [3 marks]

What is the *initial* period of the blinking LED in Listing 2?

(vi) [3 marks]

What is the *final* duty cycle of the blinking LED in Listing 2?

(vii) [2 marks]

Nate decides that he wishes to include `flasher.c` in their existing git repository, which has been initialized in his current directory and connected to a remote server. In the following box, write the git command(s) to ensure this file will be up-to-date in their remote repository.

(c) [4 marks]

What are *callgraphs* and *file dependency* diagrams? Describe how these programming aides can be used to write a software program, such as a game on the UCFK4.

END OF QUESTIONS

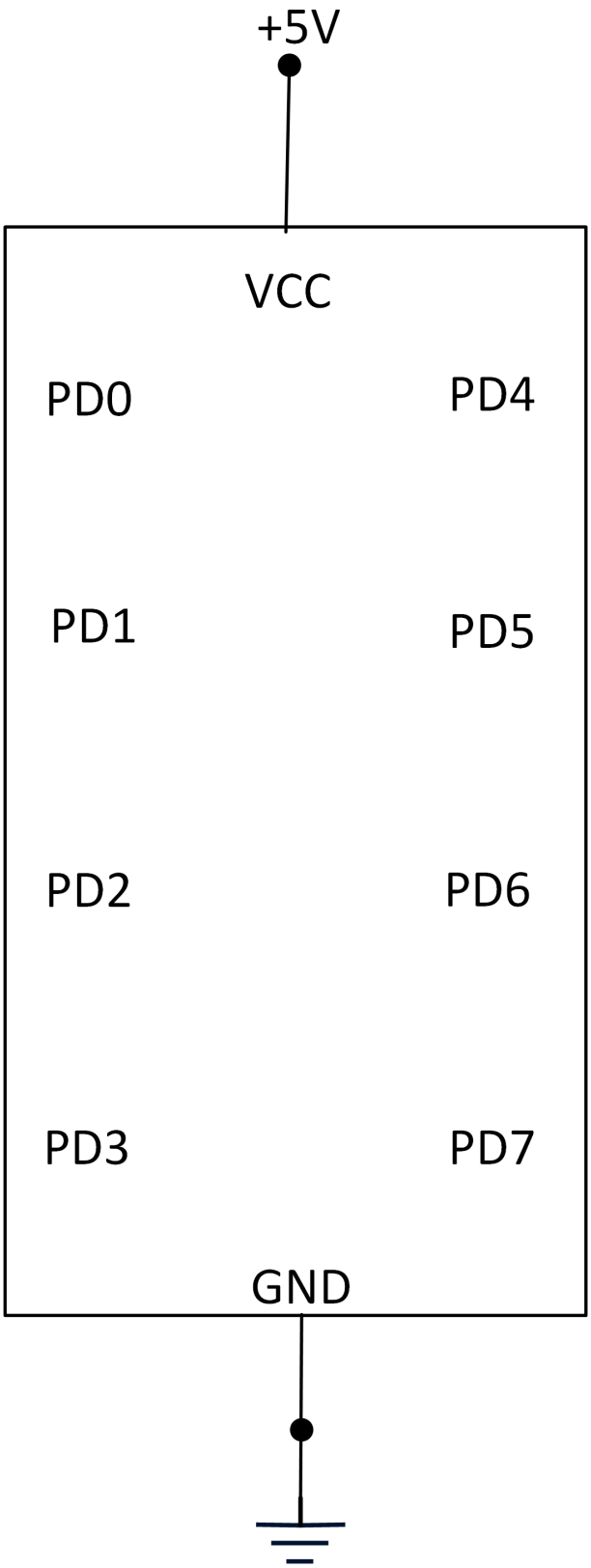


Figure 13: Extra schematic showing an ATmega32u2 microcontroller for Q3.

Extra answer space (optional)

Extra answer space (optional)

END OF EXAM